Alejandro Rodríguez del Bosque

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Education

Purdue University **Expected May 2028**
M.S. in Robotics Technology
Research: Learning-integrated control for embodied autonomous systems, emphasizing autonomous decision-making.
Relevant Coursework: Artificial Intelligence, Introduction to Robotic Systems

Tecnológico de Monterrey, Campus Monterrey **June 2026**
B.S. in Robotics and Digital Systems Engineering – 3.7 GPA
Relevant Coursework: Control Systems; System & Signal Analysis; Advanced Embedded Systems; Data Structures & Algorithms; Implementation of Intelligent Systems

Experience

Purdue University **August 2026 – Present**
Graduate Teaching Assistant

- Lead hands-on lab sessions, mentoring students through CAD, product data management, and engineering change workflows.
- Evaluate student projects and support Teamcenter-based design release, documentation, and version control.

Human Centered Robotics Laboratory, The University of Texas at Austin **August 2025 – December 2025**
Undergraduate Research Intern

- Researched learning-based control for embodied robots, focusing on stability and adaptation in dynamic environments.
- Trained RL policies in Isaac Lab and PyTorch for quadruped walking gaits that stay stable under external disturbances.
- Built and validated high-fidelity Isaac Sim environments for locomotion and navigation benchmarking.
- Designed the calibration setup, tests, and data analysis for estimating contact forces on BumpyBot, a soft-bodied robotic wheelchair, from its drivetrain torques; co-authored the resulting paper.

RASTec (IEEE RAS Student Branch), Tecnológico de Monterrey **August 2025 – June 2026**
Co-Founder & Vice President

- Co-founded the branch and led its ANYmal D research initiative, directing work on learning-based control, robust perception, and dynamic-terrain navigation; earned ANYmal Operator and Safety Certifications.
- Coordinated robotics projects using ROS 2, Gazebo, and Isaac Sim, building sim-to-real validation workflows.
- Delivered RL workshops and research training for the 2025 Latin American Robotics Symposium (LARS), covering RL implementation, experimental design, and reproducibility.

Selected Technical Projects

Autonomous Forklift Warehouse Operations (Capstone) **Spring 2026**
Developer

- Designed the system architecture and ROS 2 framework across the Jetson Nano, LiDAR, FPGA, and encoders.
- Developed PID wheel velocity control, a weighted A* planner, and Bug 2 obstacle avoidance with LiDAR-based SLAM.
- Built the perception pipeline, using YOLO logo detection and QR codes to identify and deliver pallets.
- Designed the forklift attachment in Fusion 360 and architected its FPGA-based control pipeline.

Publications

- A. Solis-Diaz, **A. Rodriguez-del-Bosque**, A. E. Ceron-Lopez, "A Pipeline for Aligning 3D Gaussian Splatting Scenes with 3D Meshes in Robotics Simulations," *2025 Latin American Robotics Symposium (LARS)*, 2025.
- J. Tomczeszyn, F. Montano, D. Zu, K. Zhang, **A. Rodriguez**, J. Jiao, L. Sentis, "In Situ Calibration and Uncertainty Quantification for Proprioceptive Wrench Estimation on Mobile Robots," in preparation.

Skills

Programming: Python, C++, C, C#, Java, Assembly
Software & Tools: ROS 2, Gazebo, NVIDIA Isaac Sim, Isaac Lab, Linux/Ubuntu, Git, Docker, Fusion 360, Teamcenter/PLM
Controls & Autonomy: PID, State Feedback, Nonlinear Control, Kinematic Modeling, State Estimation, SLAM, A*, Pure Pursuit
Machine Learning & Perception: PyTorch, Reinforcement Learning, scikit-learn, YOLOv8, OpenCV, Open3D, Gaussian Splatting
Hardware & Embedded: NVIDIA Jetson, FPGA, STM32/Arduino, ESP32, SPI, PWM, Motor Control, Sensor Integration
Languages: Spanish (Native), English (Native-level), German (Basic)